Penetration Testing Report

Footprinting & Network Scanning Phases

W2-PM-FINAL | Phase 1: Reconnaissance | Phase 2: Scanning

Field	Detail
Pentester Name (Cybersecurity Professional)	**Awais Khan**
Program / Batch	B084 – Networkwalks Cybersecurity Internship
Date of testing	8 November 2026
Report date	10 November 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client / Target	1. Networkwalks – *networkwalks.com* (written permission, see Appendix A) 2. My own local lab network – *192.168.205.0/24*
Permission secured from client?	Yes – see Appendix A
Phases covered	**Phase 1:** Reconnaissance & Footprinting **Phase 2:** Scanning & Network Discovery **Phase 3–5:** In progress

Confidential – prepared for educational purposes under written authorization

1. Liability Disclaimer

I performed these activities only on systems and devices for which I held written permission, or that I own myself. All material in this report is provided for education and research purposes only. Nothing in it may be used to break the law. The instructor, the authors and Networkwalks are not responsible for what any reader does with this knowledge, and every action a reader takes is their own responsibility. Misuse can lead to criminal charges, heavy fines, loss of employment and a permanent record. In most countries unauthorised access to a computer system is a crime even when no damage is caused.

2. Introduction

This report covers the footprinting of the *networkwalks.com* domain with several Kali Linux tools (module W2-PM1) and the scanning of my own local lab network with Zenmap (module W2-PM5). One module covers the footprinting phase and the other covers the scanning phase, so together they show how an attacker moves from collecting public information to mapping live hosts on a network. This is the Week 2 deliverable of my Networkwalks cybersecurity internship.

All footprinting commands were run from Kali Linux on 8 November 2026. The network scan was run from Zenmap on the same day against a private VMware subnet that I control. For every step this report gives the exact command, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux	Operating system used for all reconnaissance and scanning activities.
WHOIS	Find domain registration details (registrar, dates, name servers, status).
whatweb	Fingerprint web technologies (server, CMS, plugins, IP address).
nslookup	Resolve the domain name to its IP address using DNS.
curl -I	Read the HTTP response headers returned by the website.
wafw00f	Detect whether a Web Application Firewall protects the site.
dnsrecon	Enumerate DNS records (SOA, NS, MX, A, TXT/SPF, SRV).
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, IP addresses and MAC addresses, and draw the network topology.

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against *networkwalks.com* using six Kali Linux tools: WHOIS, WhatWeb, Nslookup, Curl, Wafw00f and DNSRecon. Each tool collects a different type of information about the target. Only passive and light information-gathering commands were used, as the authorization allows.

4.1.1 WHOIS – Domain registration

Command used

```
whois networkwalks.com
```

Result observed

- Registrar: **GoDaddy.com, LLC** (IANA ID 146).
- Created **6 November 2019**, last updated 12 November 2025, expires **6 November 2027**.
- Name servers: **NS6135.HOSTGATOR.COM** and **NS6136.HOSTGATOR.COM**, so DNS is hosted by HostGator.
- Domain status shows client transfer, update, delete and renew locks.
- **DNSSEC: unsigned.**

Why it matters from an attacker's point of view

WHOIS tells an attacker who manages the domain and where its DNS is hosted. The registrar and name-server provider show which companies would be targeted in a phishing or account-takeover attempt, and the dates reveal how old the domain is and when it must be renewed. Unsigned DNSSEC means DNS answers for the domain cannot be cryptographically verified.

```
┌──(root㉿kali)-[/home/kali]
└─# whois networkwalks.com
   Domain Name: NETWORKWALKS.COM
   Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois.godaddy.com
   Registrar URL: http://www.godaddy.com
   Updated Date: 2025-11-12T10:08:43Z
   Creation Date: 2019-11-06T22:51:46Z
   Registry Expiry Date: 2027-11-06T22:51:46Z
   Registrar: GoDaddy.com, LLC
   Registrar IANA ID: 146
   Registrar Abuse Contact Email: abuse@godaddy.com
   Registrar Abuse Contact Phone: 480-624-2505
   Domain Status: clientDeleteProhibited https://icann.org/epp#clientDelete
Prohibited
   Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewPr
ohibited
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTran
sferProhibited
   Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdate
Prohibited
   Name Server: NS6135.HOSTGATOR.COM
   Name Server: NS6136.HOSTGATOR.COM
   DNSSEC: unsigned
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/
wicf/
>>> Last update of whois database: 2026-11-08T11:48:28Z <<<

For more information on Whois status codes, please visit https://icann.org/
epp
```

Figure 1: WHOIS output for networkwalks.com

4.1.2 WhatWeb – Web technology fingerprint

Command used

```
whatweb networkwalks.com
```

Result observed

- HTTP redirects to HTTPS (301 Moved Permanently), then returns 200 OK.
- Web server: **Apache**. Server IP: **192.232.216.135** (United States).
- CMS: **WordPress 7.1.4**. Plugin: **WP Download Manager 3.3.76**.
- Also detected: jQuery 3.7.1, HTML5, Google Tag Manager, page title “Networkwalks Academy”, the cookie *__wpdm_client* (HttpOnly) and the public address *info@networkwalks.com*.
- WhatWeb also lists “Bootstrap 7.1.4”. That number matches the WordPress version string, so I treat this entry as unverified.

Why it matters from an attacker’s point of view

Knowing the exact CMS and plugin versions lets an attacker look up published vulnerabilities for those versions and decide whether a targeted attack is worth trying. Exposed plugin names are one of the most common starting points for WordPress attacks.

```
┌──(root㉿kali)-[/home/kali]
└─# whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTT
PServer[Apache], HttpOnly[__wpdm_client], IP[192.232.216.135], RedirectLocation[https://networkwalks.com/], Unc
ommonHeaders[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.4], Cookies[__wpdm_client], Country[UNITED STATES][US],
 Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[__wpdm_client], I
P[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph
-Protocol[website], Script[4684NR-IPIB&amp;pidnVar2=50511&amp;prtVar2=7&amp;scvVar2=12,application/json,applica
tion/ld+json,module,speculationrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions
-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.4]
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.1.4], Country[UNITED STATES][US], Email[info@networkwalk
s.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator
[WordPress 7.1.4,WordPress Download Manager 3.3.76], Open-Graph-Protocol[website], Script[4684NR-IPIB&amp;pidnV
ar2=50511&amp;prtVar2=7&amp;scvVar2=12,application/json,application/ld+json,module,speculationrules,text/javasc
ript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance
-cache-level,x-nginx-cache], WordPress[7.1.4]
```

Figure 2: WhatWeb fingerprint of networkwalks.com

4.1.3 Nslookup – DNS resolution

Command used

```
nslookup networkwalks.com
```

Result observed

- The domain resolved to **192.232.216.135** (a non-authoritative answer from my local resolver).
- This is the same IP address that WhatWeb reported.

Why it matters from an attacker’s point of view

The IP address turns a domain name into a network location. An attacker can use it for further scanning, can check which other sites share the server, and can look up the hosting provider and its IP range.



Figure 3: Nslookup result for networkwalks.com

4.1.4 Curl – HTTP response headers

Command used

```
curl -I https://networkwalks.com
```

Result observed

- Response: **HTTP/2 200**, **server: Apache**.
- The *link* header exposes the WordPress REST API root **/wp-json/** and the page endpoint */wp-json/wp/v2/pages/53*.
- *x-nginx-cache: WordPress* and *x-endurance-cache-level: 0* reveal the caching layer used by the hosting platform.
- The cookie *__wpdm_client* is set with the **secure** and **HttpOnly** flags.
- A *referrer-policy* and a *permissions-policy* are present. In the captured output I did not see Strict-Transport-Security, Content-Security-Policy, X-Frame-Options or X-Content-Type-Options.

Why it matters from an attacker's point of view

Headers confirm the software stack and show what the site does and does not enforce. The exposed REST API gives an attacker a clean, machine-readable way to enumerate pages and content, and any missing browser security headers show where client-side protections are weaker.



Figure 4: HTTP response headers returned by networkwalks.com

4.1.5 Wafw00f – Web Application Firewall detection

Command used

```
wafw00f networkwalks.com
```

Result observed

- Wafw00f v2.4.2 reported that the site is behind **ModSecurity (SpiderLabs)**.
- Detection needed only **2 requests**.

Why it matters from an attacker's point of view

Knowing the WAF product tells an attacker which filtering rules to expect and how to shape later requests. It also shows that the site owner has a protective layer in place, which is a positive finding for the defender.

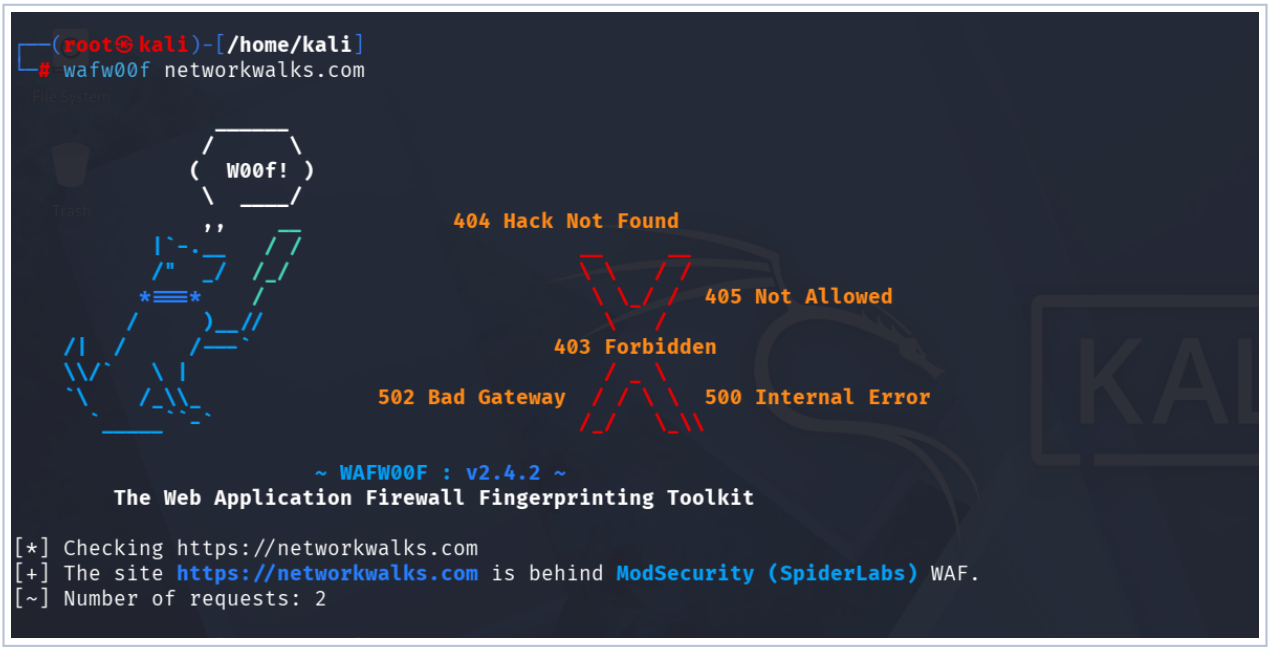


Figure 5: Wafw00f detecting ModSecurity (SpiderLabs)

4.1.6 DNSRecon – DNS enumeration

Command used

```
dnsrecon -d networkwalks.com
```

Result observed

- SOA and NS records point to **ns6135.hostgator.com** (50.87.144.87) and **ns6136.hostgator.com** (192.232.216.131).
- Both name servers disclose their software version: **BIND 9.16.23-RH**.
- Mail exchanger: **mail.networkwalks.com** at 192.232.216.135. A record: 192.232.216.135.
- TXT records include a Google site-verification entry and an **SPF** policy ending in **~all** (soft fail) that includes *websitewelcome.com*.
- Eight SRV records for *_autodiscover._tcp* point to *cpanelemaildiscovery.cpanel.net* on port 443.
- DNSRecon logged *No answer for DNSSEC query*, which fits the unsigned DNSSEC status from WHOIS.

Why it matters from an attacker's point of view

DNS data lets an attacker map the mail and name-server infrastructure, learn which hosting and email services are in use (HostGator and cPanel here) and read the SPF policy to judge how easy email spoofing might be. A disclosed BIND version can be checked against known advisories.

```
┌──(root㉿kali)-[/home/kali]
└─# dnsrecon -d networkwalks.com
2026-11-08T06:54:38.169284-0500 INFO Starting enumeration for domain: networkwalks.com
2026-11-08T06:54:38.169989-0500 INFO std: Performing General Enumeration against: networkwalks.com ...
2026-11-08T06:54:39.120415-0500 ERROR No answer for DNSSEC query for networkwalks.com
2026-11-08T06:54:40.158660-0500 INFO     SOA ns6135.hostgator.com 50.87.144.87
2026-11-08T06:54:40.852730-0500 INFO     NS ns6135.hostgator.com 50.87.144.87
2026-11-08T06:54:41.629359-0500 INFO     Bind Version for 50.87.144.87 "9.16.23-RH"
2026-11-08T06:54:41.629706-0500 INFO     NS ns6136.hostgator.com 192.232.216.131
2026-11-08T06:54:42.406172-0500 INFO     Bind Version for 192.232.216.131 "9.16.23-RH"
2026-11-08T06:54:43.832610-0500 INFO     MX mail.networkwalks.com 192.232.216.135
2026-11-08T06:54:43.848277-0500 INFO     A networkwalks.com 192.232.216.135
2026-11-08T06:54:45.061890-0500 INFO     TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4
q75X-Ij-mjgEeg-UsYI
2026-11-08T06:54:45.062656-0500 INFO     TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitew
elcome.com ~all
2026-11-08T06:54:45.792978-0500 INFO Enumerating SRV Records
2026-11-08T06:54:51.213581-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.ne
t 184.94.203.11 443
2026-11-08T06:54:51.214338-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.ne
t 184.94.203.15 443
2026-11-08T06:54:51.214536-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.ne
t 184.94.204.11 443
2026-11-08T06:54:51.214688-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.ne
t 184.94.204.15 443
2026-11-08T06:54:51.214854-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.ne
t 184.94.204.14 443
2026-11-08T06:54:51.215917-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.ne
t 184.94.203.9 443
2026-11-08T06:54:51.216098-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.ne
t 184.94.203.14 443
2026-11-08T06:54:51.216283-0500 INFO     SRV _autodiscover._tcp.networkwalks.com cpanelemaildiscovery.cpanel.ne
t 184.94.204.9 443
2026-11-08T06:54:51.220565-0500 INFO 8 Records Found
```

Figure 6: DNSRecon enumeration of networkwalks.com

4.2 Network Scanning with Zenmap

For the second activity I used **Zenmap** to discover hosts on my own local lab network, which is a private VMware subnet. The task was to find the local subnet, discover live hosts, record their IP and MAC addresses, and generate a network topology.

Command used

```
nmap -sn 192.168.205.131/24
```

I entered the target *192.168.205.131/24* in Zenmap and selected the **Ping scan** profile, which runs *nmap -sn*. This is a host-discovery scan only: it checks which addresses answer, and it does not scan ports. The scan covered **256 IP addresses** and found **6 hosts up** in **2.96 seconds**.

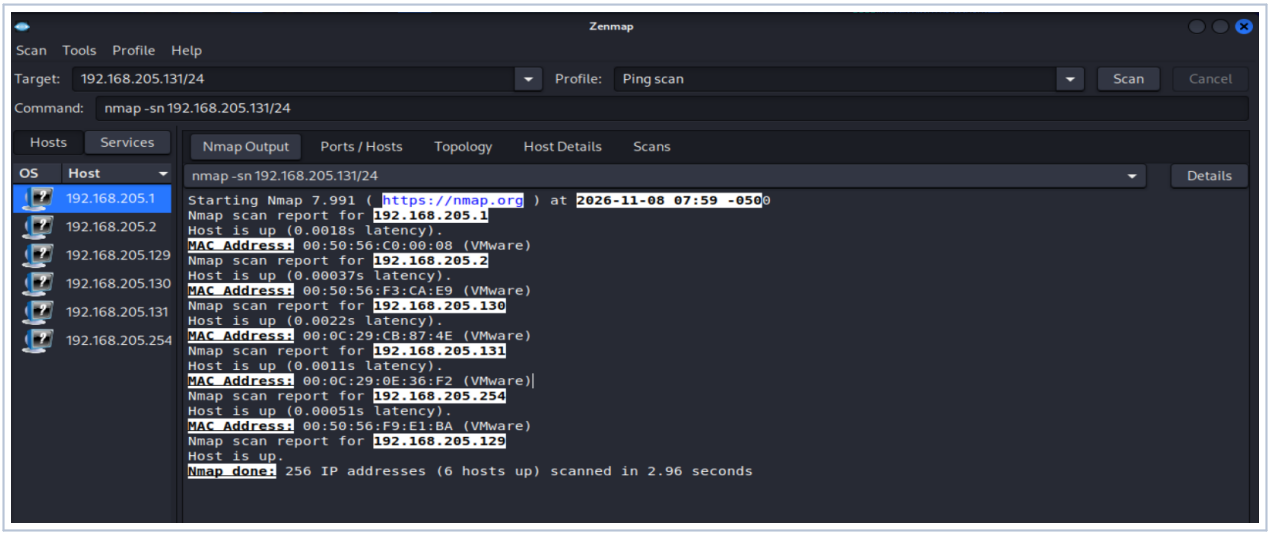


Figure 7: Zenmap ping scan: Nmap output listing six live hosts

Live hosts discovered

#	IP address	MAC address	Vendor	Latency
1	192.168.205.1	00:50:56:C0:00:08	VMware	0.0018 s
2	192.168.205.2	00:50:56:F3:CA:E9	VMware	0.00037 s
3	192.168.205.130	00:0C:29:CB:87:4E	VMware	0.0022 s
4	192.168.205.131	00:0C:29:0E:36:F2	VMware	0.0011 s
5	192.168.205.254	00:50:56:F9:E1:BA	VMware	0.00051 s
6	192.168.205.129	Not displayed	–	–

All five reported MAC addresses belong to VMware, which confirms that the subnet is a virtual lab network. Nmap does not print a MAC address for the machine it runs on, which is why 192.168.205.129 has none.

Network topology

After the scan I opened the **Topology** tab in Zenmap. It shows the scanning machine as *localhost* in the centre, with the six discovered hosts arranged around it. I used the Legend and Save Graphic controls to export the topology as required by the practical.

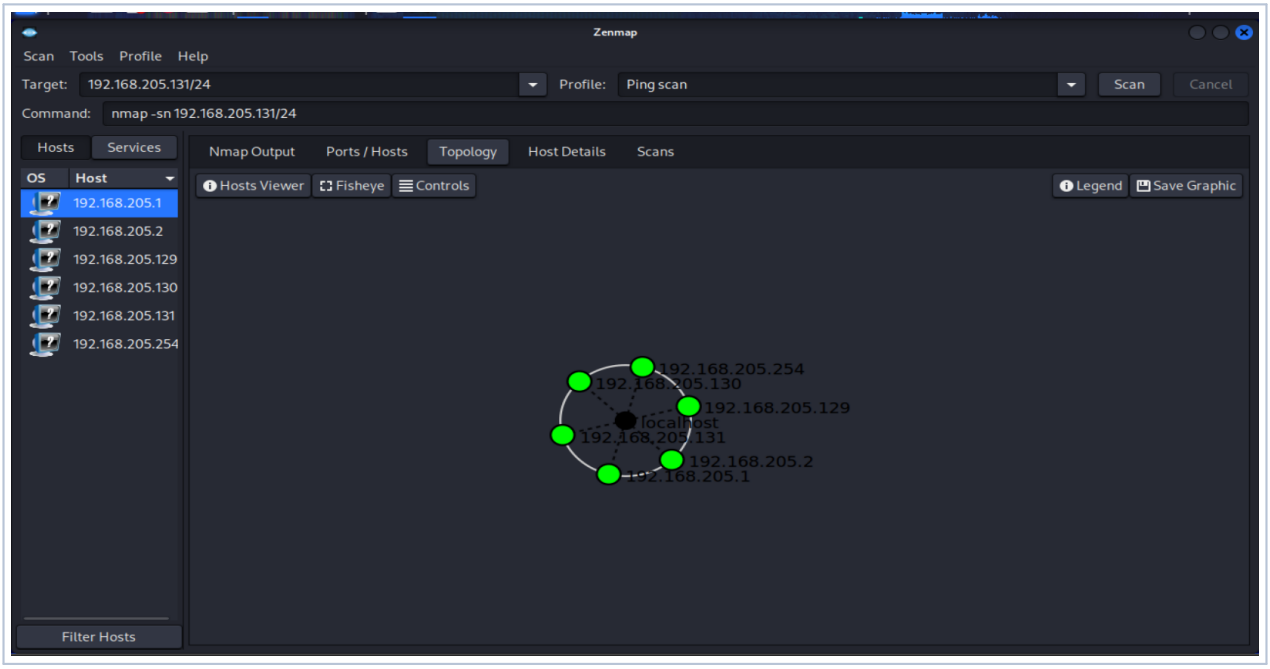


Figure 8: Zenmap topology view of the discovered hosts

Why it matters from an attacker's point of view

Host discovery is the first step of internal reconnaissance. It shows which devices are alive, how they are addressed and what hardware or virtualisation they run on. From this list an attacker would choose which hosts to port-scan next. For a defender, the same scan is a simple way to confirm that only expected devices are on the network.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	**Web technology and version information exposed**	WhatWeb showed WordPress 7.1.4 and WP Download Manager 3.3.76	Attackers can search for known vulnerabilities in these exact versions	**Medium**
2	**Server IP address identifiable**	Nslookup resolved the domain to 192.232.216.135	Reveals the network location of the web and mail service	**Low**
3	**HTTP technical information exposed**	Curl showed Apache, caching headers and the /wp-json/ REST API link	Helps technology fingerprinting and content enumeration	**Low**
4	**Browser security headers not observed**	No HSTS, CSP, X-Frame-Options or X-Content-Type-Options in the captured headers	Weaker client-side protection against clickjacking and content-type attacks	**Low**
5	**WAF technology identifiable**	Wafw00f identified ModSecurity (SpiderLabs) in 2 requests	Reveals part of the security architecture (the WAF itself is a positive control)	**Low**
6	**DNS records and name-server software version exposed**	DNSRecon listed NS, MX, SRV and TXT records and BIND 9.16.23-RH	Helps build an infrastructure profile; the version can be checked against advisories	**Medium**
7	**DNSSEC not enabled and SPF ends in soft fail**	WHOIS showed DNSSEC unsigned; the SPF record ends with ~all	DNS answers cannot be verified and email spoofing is easier to attempt	**Low**
8	**Public contact email visible**	WhatWeb found info@networkwalks.com on the home page	Can be used as a target for phishing or spam	**Low**
9	**Multiple live hosts visible on the lab subnet**	Zenmap found 6 live hosts, all with VMware MAC addresses	Unknown devices on a network could go unnoticed (expected in my own lab)	**Low**

Risk level key: **Critical** **Medium** **Low**

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities.

These exercises involved information gathering and host discovery only. No exploitation or vulnerability validation was performed. Seeing a software version, an IP address or a DNS record does not by itself mean that a system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on these observations, I recommend the following security improvements:

1. Reduce exposed technology information

Hide or remove WordPress and plugin version strings from page output and review what the site reveals about its CMS and plugins.

2. Keep software updated

Update WordPress core, the WP Download Manager plugin and all other components regularly, and review them against current security advisories.

3. Restrict unnecessary REST API exposure

Review whether the public /wp-json/ endpoints are needed and limit anonymous access where possible.

4. Add browser security headers

Review the HTTP headers and add Strict-Transport-Security, Content-Security-Policy, X-Frame-Options and X-Content-Type-Options where suitable.

5. Review DNS configuration

Ask the DNS provider to hide the BIND version, consider enabling DNSSEC and check that only required records are public.

6. Strengthen email authentication

Review the SPF policy (consider a stricter -all once all senders are listed) and add DKIM and DMARC if they are not already in place.

7. Keep the WAF enabled and tuned

Keep ModSecurity enabled with up-to-date rules, since it already blocks basic attacks.

8. Perform regular internal network discovery

Scan owned networks periodically, investigate any unknown device, and keep network topology and device documentation up to date.

9. Test only with authorization

Reconnaissance and scanning should only be performed against systems and networks where written authorization exists.

7. Conclusion

During Week 2 of my Cybersecurity internship I completed practical activities in footprinting, reconnaissance and network scanning.

In the footprinting activity I used six Kali Linux tools to collect information about the target domain. WHOIS gave the registration and name-server details, WhatWeb identified the web technologies, Nslookup resolved the domain to its IP address, Curl exposed the HTTP headers and the REST API link, Wafw00f identified the ModSecurity WAF, and DNSRecon mapped the DNS, mail and service records.

In the network scanning activity I used Zenmap to run a ping scan of my own lab subnet. It found six live hosts, showed their IP and MAC addresses, and produced a network topology.

These exercises showed me that a security professional can learn a great deal about an environment before trying to exploit anything, simply by analysing public information and network responses. They also taught me that technical findings must be documented clearly: a good report explains what was done, what was found, what it means, what risk it creates and how that risk can be reduced.

Finally, reconnaissance and scanning must always stay within an authorized scope. All activities in this report were completed as part of the assigned educational cybersecurity lab.

8. Evidences Collected

Each screenshot is shown in Section 4 next to the activity it supports. The table below indexes them.

Figure	Tool	Command / Action	Captured (as shown on screen)
1	WHOIS	whois networkwalks.com	Database update 2026-11-08 11:48 UTC
2	WhatWeb	whatweb networkwalks.com	8 Nov 2026
3	Nslookup	nslookup networkwalks.com	8 Nov 2026
4	Curl	curl -I https://networkwalks.com	Header date: Sun, 08 Nov 2026 11:52:23 GMT
5	Wafw00f	wafw00f networkwalks.com	8 Nov 2026
6	DNSRecon	dnsrecon -d networkwalks.com	2026-11-08 06:54 (UTC-5)
7	Zenmap	nmap -sn 192.168.205.131/24	2026-11-08 07:59 (UTC-5)
8	Zenmap	Topology tab	8 Nov 2026

Appendix A: Letter of Authorization

The written permission from Networkwalks that covers this testing is attached below.

[Signed Letter of Authorization to be inserted here]

The letter must name the tester, be signed by Networkwalks and cover 8 November 2026.

End of report